

EECS 489

Computer Networks

Winter 2025

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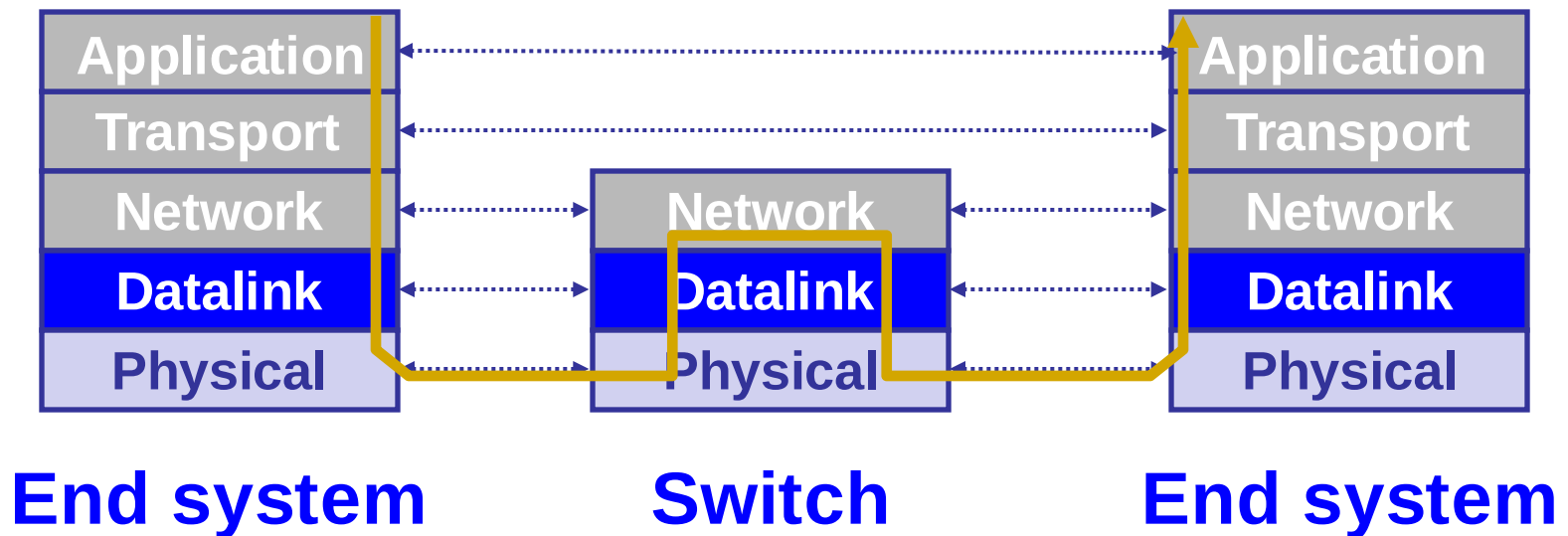
Material with thanks to Aditya Akella, Sugih Jamin, Philip Levis, Sylvia Ratnasamy, Peter Steenkiste, and many other colleagues.

Agenda

- Data link layer

Data link layer

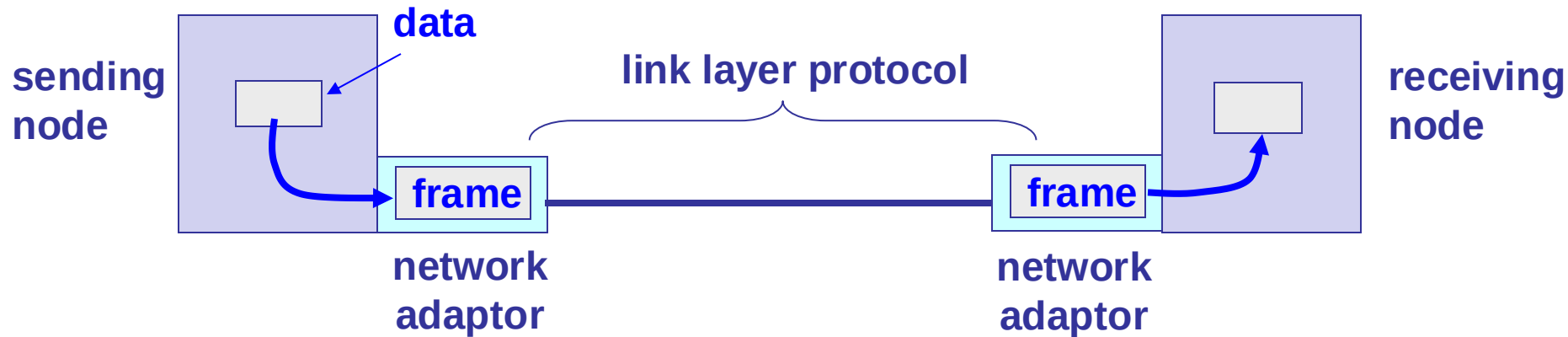
- Present everywhere
- Transfers data between adjacent nodes or between nodes on the same local area network



Data link layer

- Provides four primary services
 - Framing
 - » Encapsulates network layer data
 - Link access
 - » Medium access control (MAC) protocol defines when to transmit frames
 - Reliable delivery
 - » Primarily for mediums with high error rates (e.g., wireless)
 - Error detection and correction

Packets are now “frames”



- ❑ Frames encapsulate network layer packets
- ❑ Link layer protocols are implemented in h/w
- ❑ Frame formats can change based on link layer protocol

Point-to-point vs. broadcast medium

- **Point-to-point**: dedicated pairwise communication
 - E.g., long-distance fiber link
 - E.g., Point-to-point link b/n Ethernet switch and host
- **Broadcast**: shared wire or medium
 - Traditional Ethernet (pre ~2000)
 - 802.11 wireless LAN

Multiple access algorithm

- Context: a shared broadcast channel
 - Must avoid having multiple nodes speaking at once
 - » Otherwise, collisions lead to garbled data
 - Need distributed algorithm to determine which node can transmit
- Three classes of techniques
 - **Channel partitioning**: divide channel into pieces
 - **Taking turns**: scheme for deciding who transmits
 - **Random access**: allow collisions, and then recover
 - » More in the Internet style!

Random access MAC protocols

- ❑ When node has packet to send
 - Transmit at full channel data rate **w/o** coordination
- ❑ Two or more transmitting nodes $\Rightarrow$ **collision**
 - Data lost
- ❑ Random access MAC protocol specifies
 - How to **detect** and **recover** from collisions
- ❑ Examples
 - ALOHA and Slotted ALOHA
 - **CSMA**, **CSMA/CD**, CSMA/CA (wireless)

Ethernet

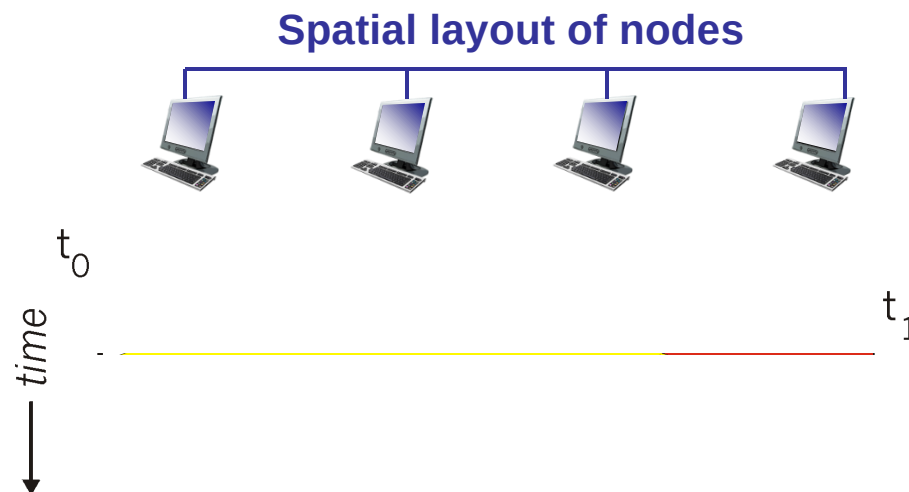
- ❑ Invented as a broadcast technology
 - Hosts share channel
 - Each packet received by all attached hosts
 - CSMA/CD for media access control
- ❑ Modern Ethernets are “switched” (later)
 - Point-to-point links between switches and between a host and switch
 - No sharing $\Rightarrow$ no CSMA/CD
 - » Uses “self learning” and “spanning tree” algorithms for routing

CSMA (Carrier Sense Multiple Access)

- ❑ CSMA: listen before transmit
 - If channel sensed idle: transmit entire frame
 - If channel sensed busy, defer transmission
- ❑ Human analogy: don't interrupt others!
- ❑ Does not eliminate all collisions
 - Why?

CSMA collisions

- ❓ **Propagation delay:** two nodes may not hear each other before sending
- ❓ CSMA reduces but does not eliminate collisions
- ❓ Collision: entire packet transmission time wasted
 - Distance and propagation delay affect collision probability

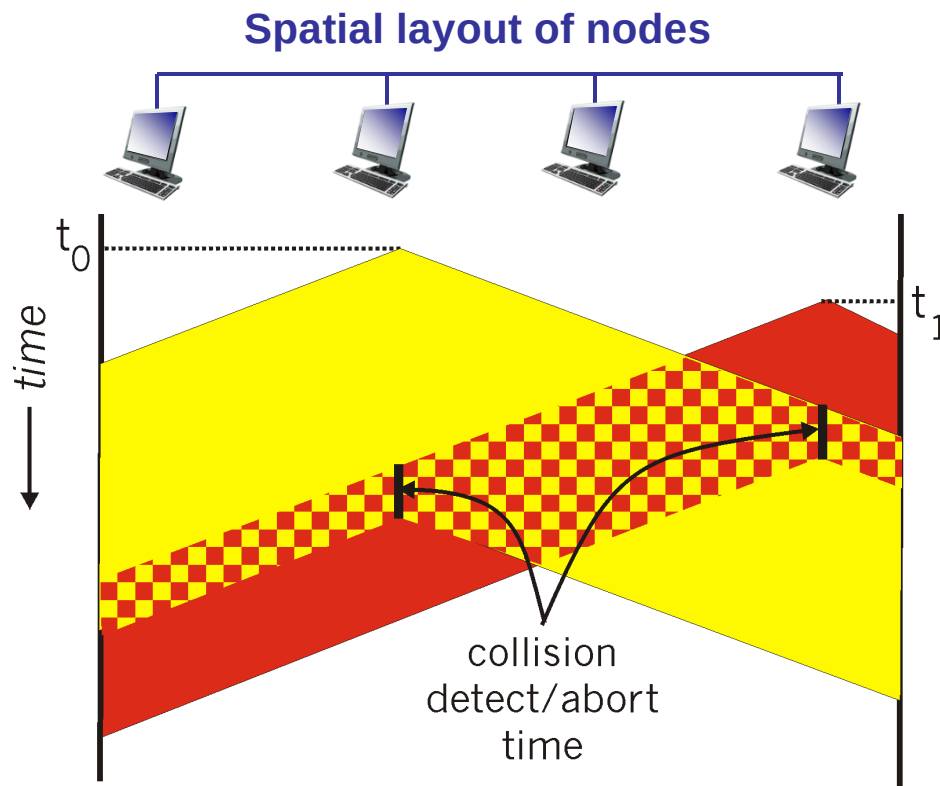


CSMA/CD (Collision Detection)

- CSMA/CD: carrier sensing, deferral as in CSMA
 - Collisions detected within short time
 - Colliding transmissions aborted, reducing wastage
- Collision detection easy in wired (broadcast) LANs
 - Compare transmitted, received signals
- Collision detection difficult in wireless LANs
 - Later!

CSMA/CD (Collision Detection)

- For this to work, need restrictions on minimum frame size and maximum distance
 - Why?



Limits on CSMA/CD network length



- ❑ Latency depends on physical length of link
 - Time to propagate a frame from one end to other
- ❑ Suppose A sends a frame at time t
 - And B sees an idle line at a time just before $t + d$
 - ... so B happily starts transmitting a frame
- ❑ B detects a collision, and sends jamming signal
 - But A cannot see collision until $t + 2d$

Limits on CSMA/CD network length



- ❑ A needs to wait for time **$2d$** to detect collision
 - So, A should keep transmitting during this period
 - AND keep an eye out for a possible collision
- ❑ Imposes restrictions; e.g., for 10 Mbps Ethernet
 - **Maximum length** of the wire: 2,500 meters
 - **Minimum length** of a frame: 512 bits (64 bytes)

Three key ideas of random access

□ Carrier sense

- Listen before speaking and don't interrupt
- Checking if someone else is already sending data
- ... and waiting till the other node is done

□ Collision detection

- If someone else starts talking at the same time, stop
 - » Make sure everyone knows there was a collision!
- Realizing when two nodes are transmitting at once
- ...by detecting that the data on the wire is garbled

Three key ideas of random access

□ Randomness

- Don't start talking again right away
- Waiting for a random time before trying again

How long should you wait?

- Should it be immediate?
- Should it be a random number with a fixed distribution?

Ethernet: CSMA/CD Protocol

- ❑ Carrier sense: wait for link to be idle
- ❑ Collision detection: listen while transmitting
 - No collision: transmission is complete
 - Collision: abort transmission & send jam signal
- ❑ Random access: **binary exponential back-off**
 - After collision, wait a random time before retrying
 - After m^{th} collision, choose K randomly from $\{0, \dots, 2^m - 1\}$
 - » Wait for $K \cdot 512$ bit times before trying again
 - » If transmission occurring when ready to send, wait until end of transmission (CSMA)

Efficiency of CSMA/CD

- Efficiency is defined as the long-run fraction of time during which frames are being transmitted without collision
- d_{prop} = max propagation time between two adapters
- d_{trans} = time to transmit a max-sized frame

$$\text{Efficiency} \approx \frac{1}{1 + 5 d_{\text{prop}} / d_{\text{trans}}}$$

Efficiency of CSMA/CD

□ $d_{\text{prop}} \rightarrow 0$

- Efficiency approaches 1
- Colliding nodes abort immediately

□ $d_{\text{trans}} \rightarrow \infty$

- Efficiency approaches 1
- Each frames uses the channel for a long time

$$\text{Efficiency} \approx \frac{d_{\text{trans}}}{d_{\text{trans}} + 5 d_{\text{prop}}}$$

5-MINUTE BREAK!

Announcements

- Assignment 4 is out!

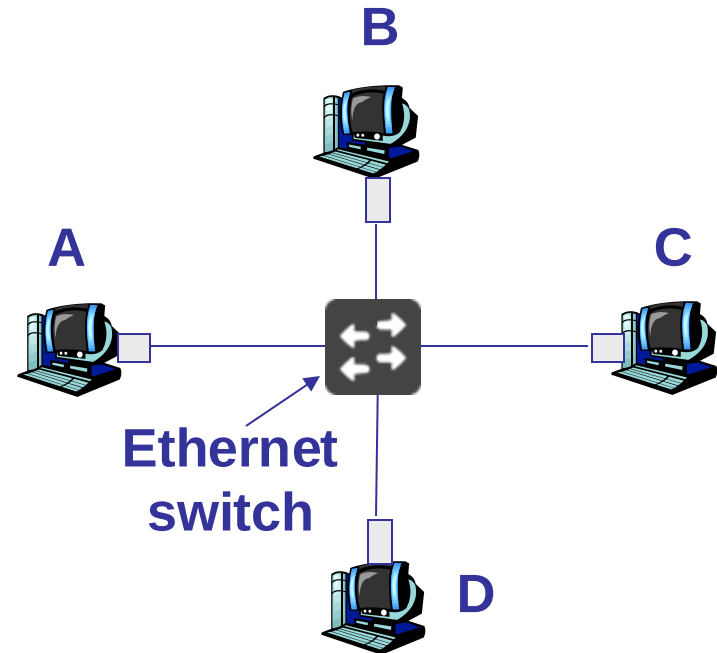
SWITCHED ETHERNET

Broadcast vs. switched Ethernet

- ❑ Invented as a broadcast technology
 - Hosts share channel
 - Each packet received by all attached hosts
 - CSMA/CD for media access control
- ❑ **Modern Ethernets are “switched”**
 - Point-to-point links between switches and between a host and switch
 - No sharing $\Rightarrow$ no CSMA/CD
 - » Uses “self learning” and “spanning tree” algorithms for routing

Why switched Ethernet?

- ▣ Enables concurrent communication
 - Host A can talk to C, while B talks to D
 - No collisions and no need for CSMA/CD
 - No constraints on link lengths, etc.



The evolution of Ethernet

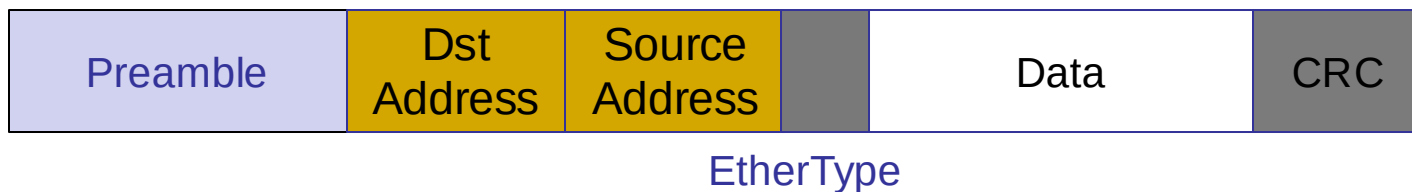
- Changed almost everything except the frame format
 - From the shared media coax cables to dedicated links
 - From 3 Mbit/s to 100 Gbit/s
 - From electrical signaling to optical
- **Lesson:** the right interface can accommodate many changes
 - Evolve the implementation while maintaining the interface (backward compatibility)

Topics

- ▣ Frames and framing
- ▣ Addressing
- ▣ Routing
- ▣ Forwarding
- ▣ Discovery

Ethernet “Frames”

- ❑ Encapsulates IP datagram



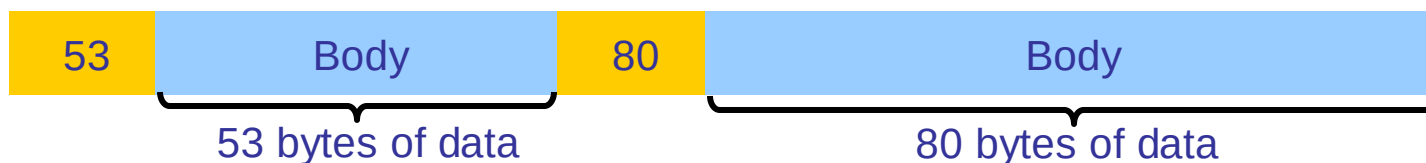
- ❑ **Preamble**: 7 bytes for clock synchronization and 1 byte to indicate start of frame
- ❑ **Addresses**: 6 bytes each
- ❑ **EtherType**: 2 bytes, protocol (e.g., IP), length, ..
- ❑ **Data payload**: max 1500 bytes, min 46 bytes
- ❑ **CRC**: 4 bytes for error detection

Framing frames

- Physical layer puts bits on a link
- But, two hosts connected on the same physical medium need to be able to exchange frames
 - Service provided by the link layer
 - Implemented by the network adaptor
- **Framing problem**: how does the link layer determine where each frame begins and ends?

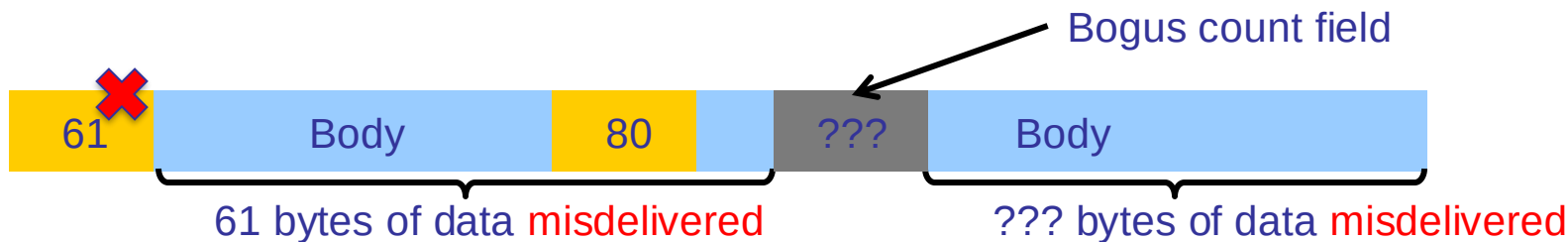
Simple approach: Count bytes

- Sender includes number of bytes in header



- Receiver extracts this number of bytes of body

- What if the Count field is corrupted?



- L2 will frame the wrong bytes → a framing error
- CRC tells you to discard this frame, but what about the next one?

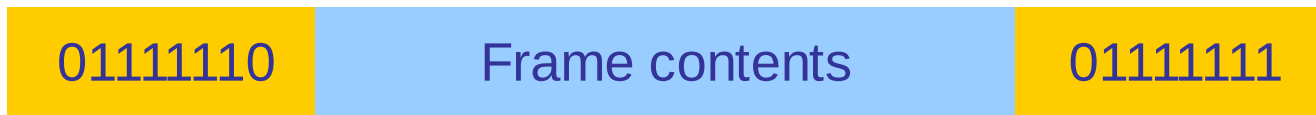
Desynchronization

- Once framing on a link is desynchronized, it can stay that way
- Need a method to **resynchronize**

Framing with sentinel bits

- Delineate frame with special “sentinel” bit pattern

- e.g., 01111110 $\Rightarrow$ start, 01111111 $\Rightarrow$ end



- What if sentinel occurs within frame?
- Solution: **bit stuffing**
 - Sender always inserts a 0 after five 1s in the frame contents
 - Receiver always removes a 0 appearing after five 1s

When receiver sees five 1s...

01111110

Frame content

01111111

- ❑ If next bit 0, remove it; begin counting again
 - Because this must be a stuffed bit; we can't be at beginning/end of frame (those had six or seven 1s)
- ❑ If next bit 1 (i.e., we've seen six 1s) then:
 - If following bit is 0, this is start of frame
 - » Because the receiver has seen 01111110
 - If following bit is 1, this is end of frame
 - » Because the receiver has seen 01111111

Example: sentinel bits

- ❑ Original data, including start/end of frame:
 - 01111110011111101111101111100101111111
- ❑ Sender rule: five 1s → insert a 0
 - After bit stuffing at the sender:
 - 01111110011111010111110011111000101111111
- ❑ Receiver rule: five 1s and next bit 0 → remove 0
 - 01111110011111101111101111100101111111

Topics

- ▣ Frames and framing
- ▣ Addressing
- ▣ Routing
- ▣ Forwarding
- ▣ Discovery

Medium Access Control (MAC) Address

❑ MAC address

- Numerical address associated with a network adapter
- Flat name space of 48 bits (e.g., 00-15-C5-49-04-A9 in HEX)
- Unique, hard-coded in the adapter when it is built

❑ Hierarchical Allocation

- **Blocks**: assigned to vendors (e.g., Dell) by the IEEE
 - » First 24 bits (e.g., 00-15-C5-**-**-**)
- **Adapter**: assigned by the vendor from its block
 - » Last 24 bits

MAC address vs. IP address

MAC Addresses

- ❑ Hard-coded when adapter is built
- ❑ Flat name space of 48 bits (e.g., 00-0E-9B-6E-49-76)
- ❑ Like a social security number
- ❑ Portable, and can stay the same as the host moves
- ❑ Used to get packet between interfaces on same network

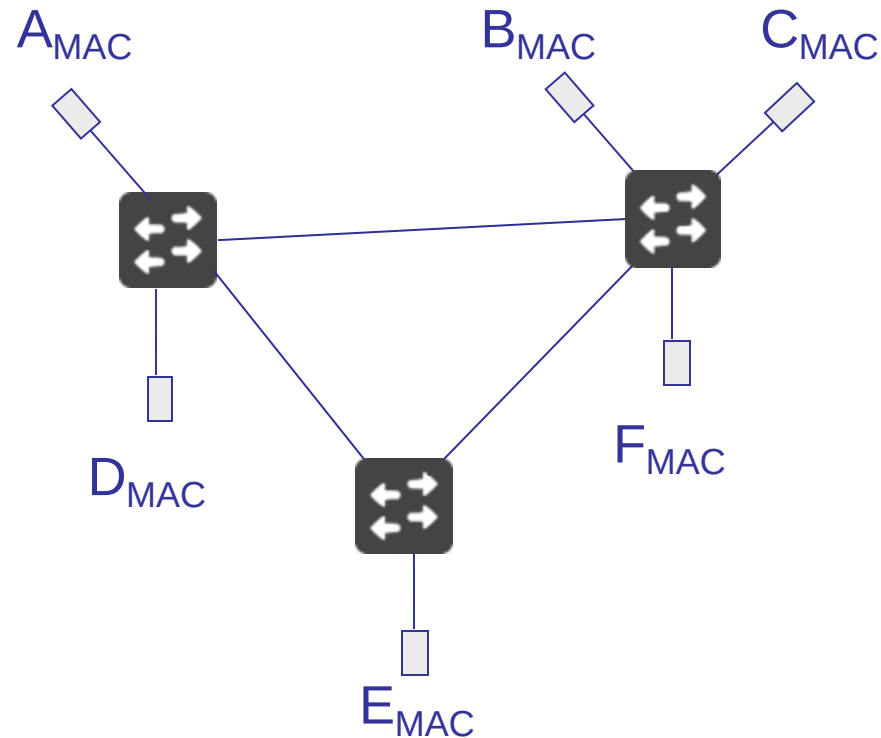
IP Addresses

- ❑ Configured, or learned dynamically
- ❑ Hierarchical name space of 32 bits for IPv4 (e.g., 12.178.66.9)
- ❑ Like a postal mailing address
- ❑ Not portable, and depends on where the host is attached
- ❑ Used to get a packet to destination IP subnet

Topics

- ▣ Frames and framing
- ▣ Addressing
- ▣ Routing
- ▣ Forwarding
- ▣ Discovery

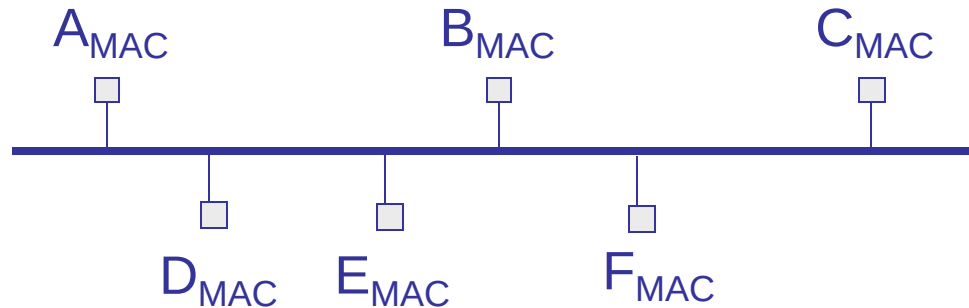
Routing with switched Ethernet?



Why does Ethernet not use LS/DV?

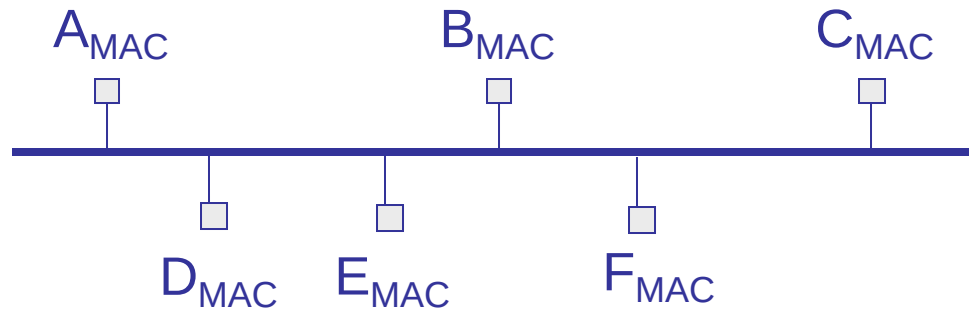
- ❑ Concerns over scalability
 - Flat MAC addresses cannot be aggregated like IP addresses
- ❑ Legacy

“Routing” with broadcast Ethernet



- ❑ Sender transmits frame onto broadcast link
- ❑ Each receiver's link layer passes the frame to the network layer:
 - If destination address matches the receiver's MAC address OR if the destination address is the broadcast MAC address (ff:ff:ff:ff:ff:ff)

“Routing” with broadcast Ethernet

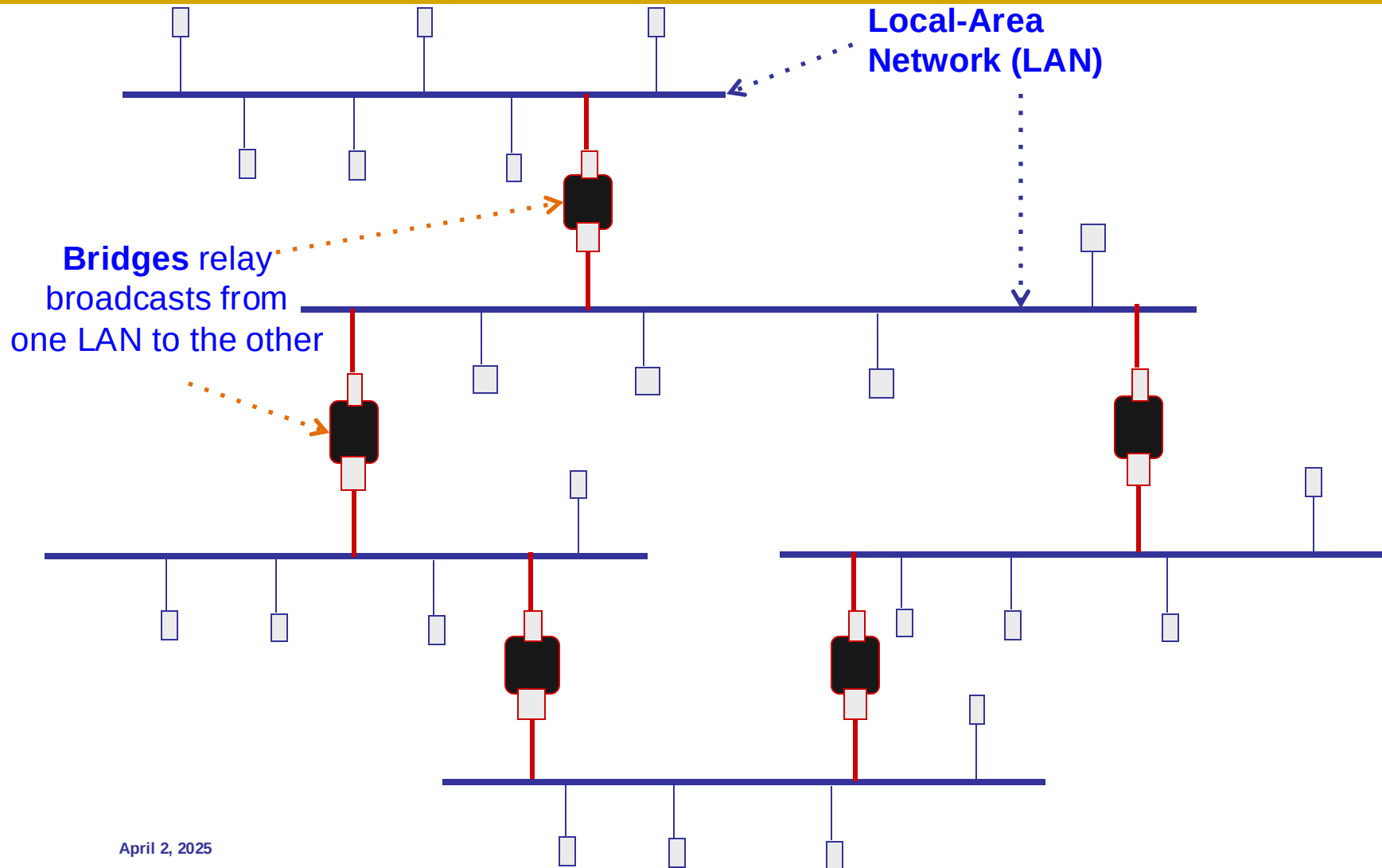


- ❑ Ethernet is “plug-n-play”
- ❑ A new host plugs into the Ethernet and is good to go
 - No configuration by users or network operators
 - Broadcast as a means of bootstrapping communication

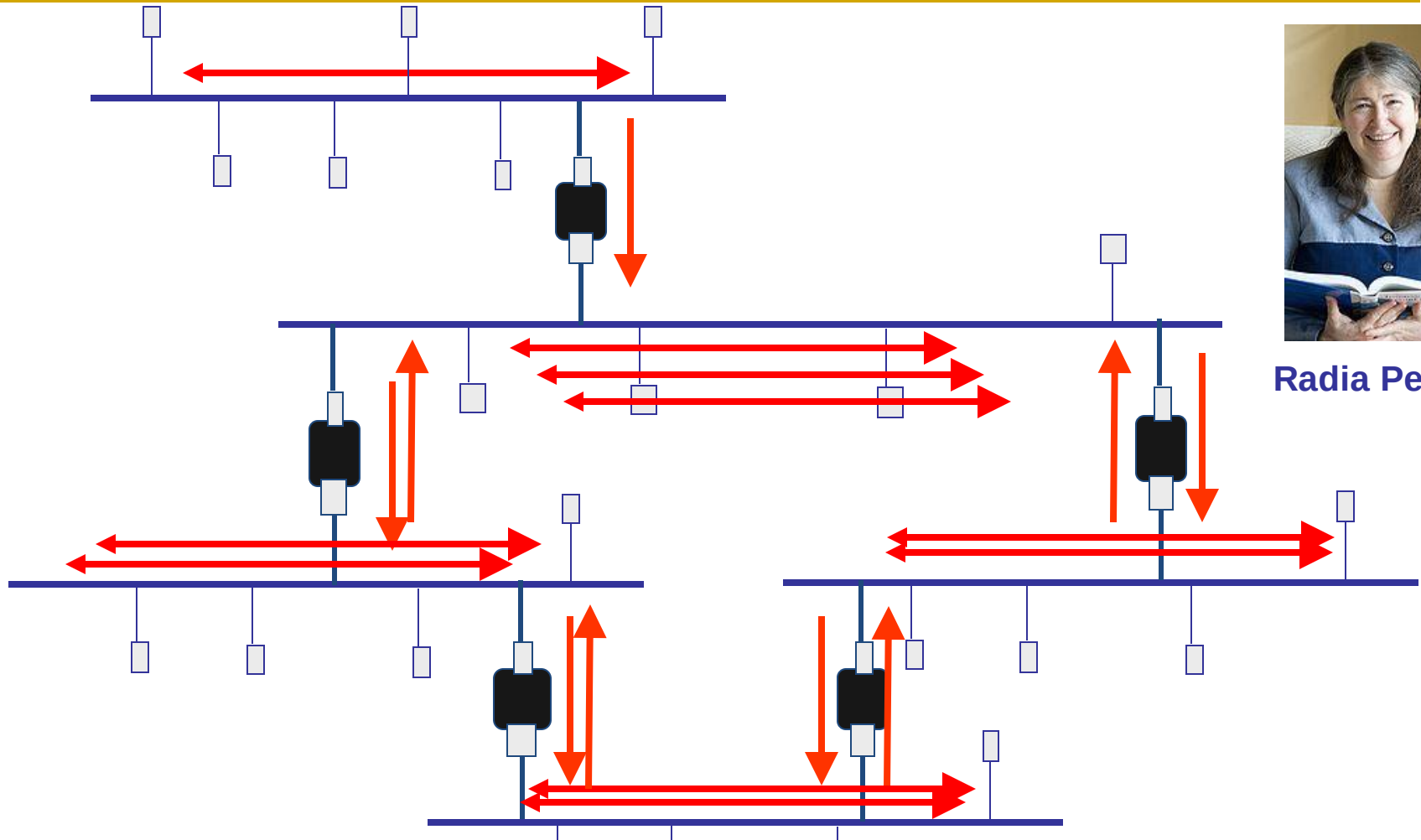
Why does Ethernet not use LS/DV?

- ❑ Concerns over scalability
 - Flat MAC addresses cannot be aggregated like IP addresses
- ❑ Legacy
 - Backward compatibility with broadcast Ethernet
 - Desire to maintain Ethernet's plug-n-play behavior
 - How broadcast Ethernet evolved

Routing in extended LANs



The “broadcast storm” problem



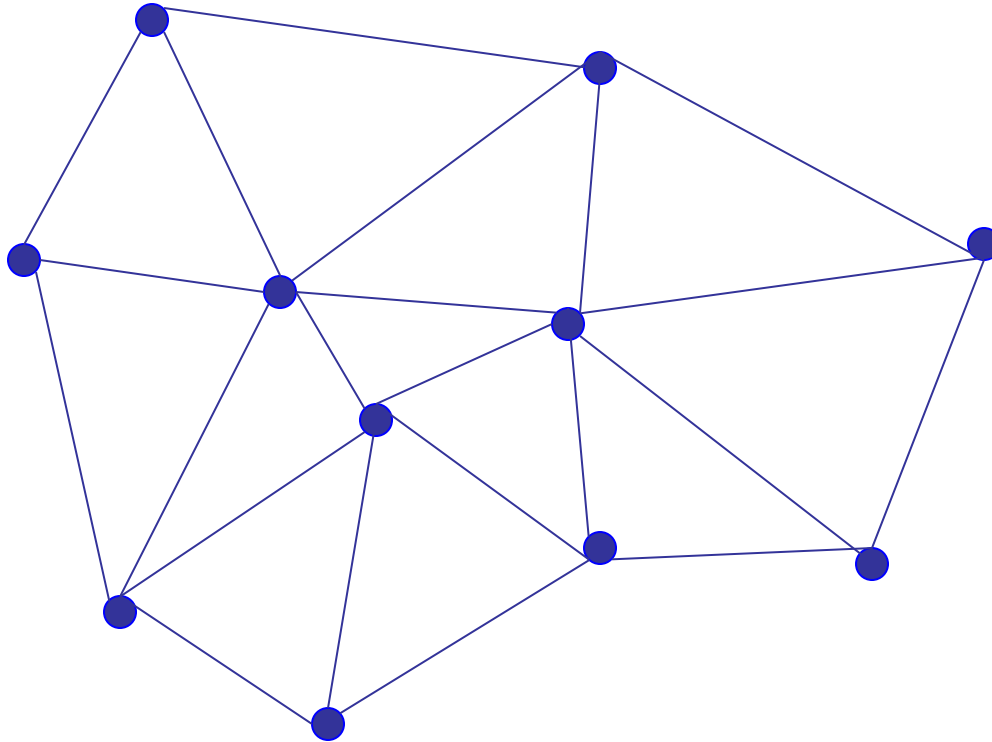
Radia Perlman

Perlman's idea: eliminate loops in the topology

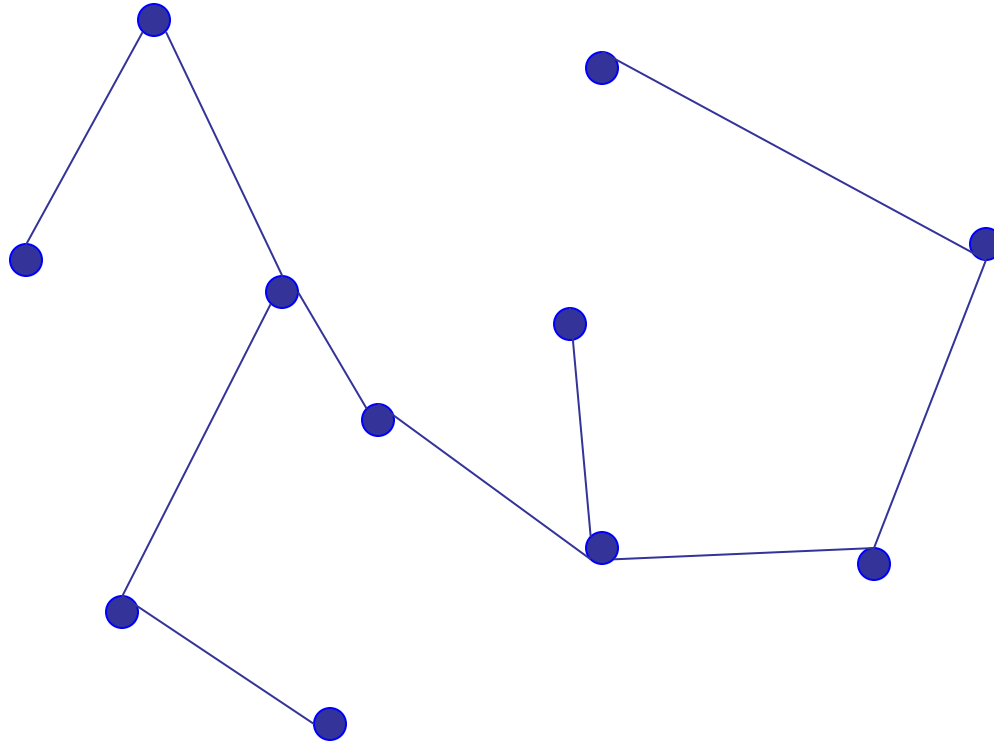
Easiest way to avoid loops

- Use a topology where loops are impossible!
- Take arbitrary topology and build a **spanning tree**
 - Sub-graph that includes all vertices but contains no cycles
 - Links not in the spanning tree are not used to forward frames

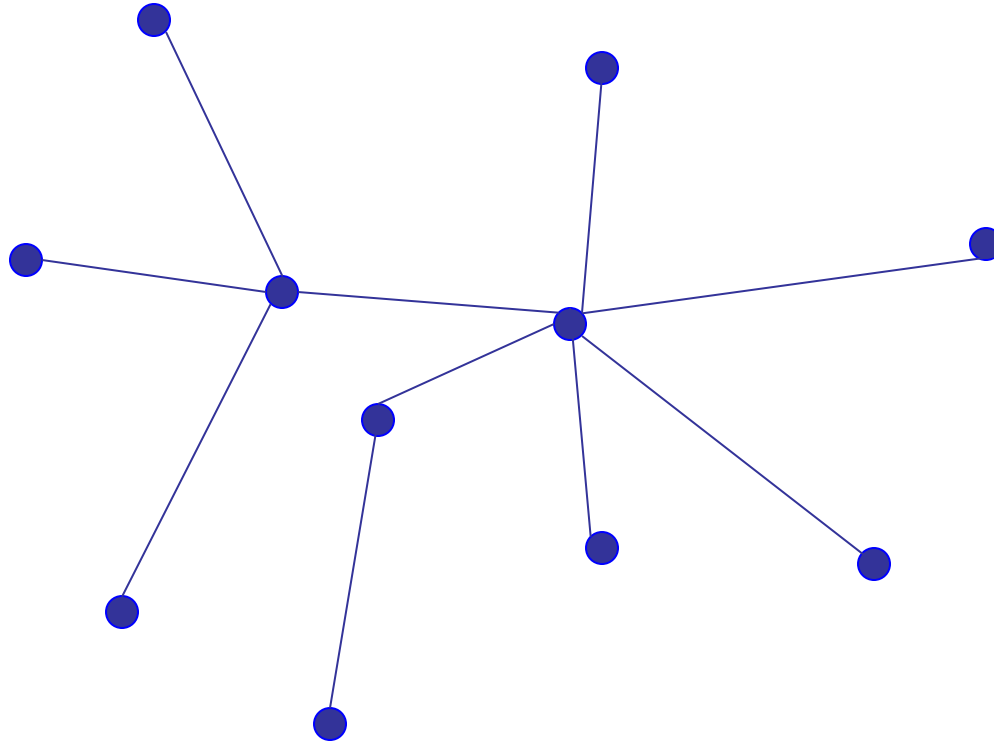
Consider a graph



A spanning tree



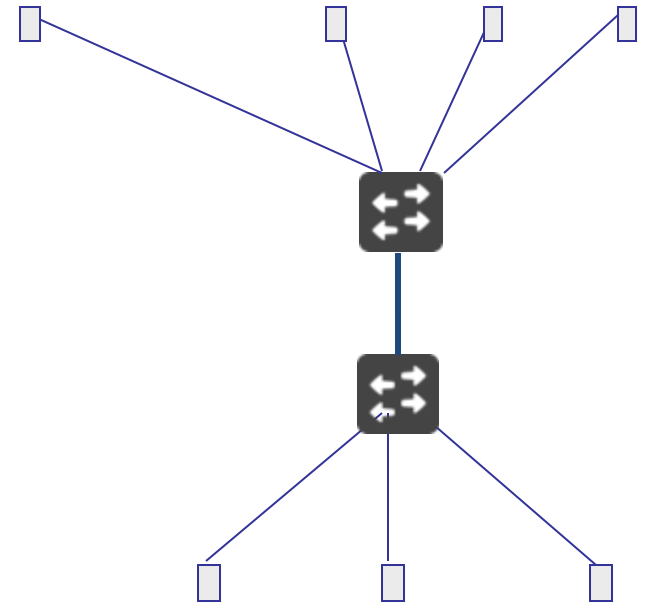
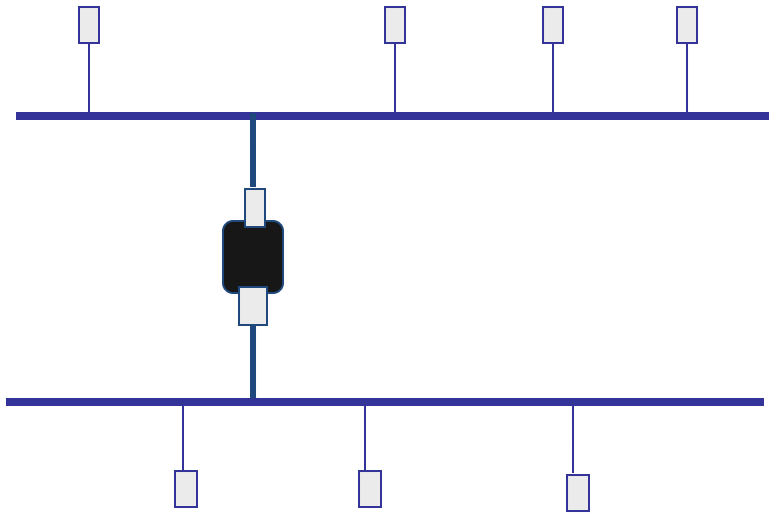
Yet another spanning tree



Spanning tree protocol (Perlman'85)

- ❑ Protocol by which bridges construct a spanning tree
- ❑ Nice properties
 - Zero configuration (by operators or users)
 - Self healing
- ❑ Still used today

From extended LANs to switched Ethernet



Switched Ethernet

- Constraints (for backward compatibility)
 - No changes to end-hosts
 - Maintain plug-n-play aspect
- Earlier Ethernet achieved plug-n-play by leveraging a broadcast medium
 - Can we do the same in a switched topology?

Summary

- Data link layer transfers data between adjacent nodes or nodes connected to the same switch
- Ethernet evolved from a broadcast medium to switched
- **Next week:** Link layer wrap up + putting everything together